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Metals & Non-Metals

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Enjoy yourselves!

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Metals are elements that conduct heat and electricity and are malleable and ductile. • All metals → Solid (except mercury) • Metals are electropositive. • Metals are the elements (except hydrogen) which form +ve ions by losing electrons. • Most abundant metal in Earth's crust is Al. Non-Metals They do not conduct heat and electricity and are neither malleable nor ductile. • Out of 22 non-metals, 10 are solids, 1(bromine) is a liquid whereas remaining 11 are gases. • Non-metals are electronegative. • H is only non-metal which loses electron. Metals ① They are malleable and ductile, and can be beaten and drawn into thin sheets and wires. ② They are good conductors of heat and electricity. ③ They are lustrous and can be polished. ④ They are hard (except sodium and potassium). Non-Metals ① Neither malleable nor ductile. ② They do not conduct heat and electricity (except graphite). ③ They are not lustrous. ④ They are generally soft (except diamond).

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⑤ They have high Melting and Boiling point ⑤ They have comparatively low Melting and (except Sodium and Potassium). ⑥ They are Sonorous. ⑥ They are non-sonorous. Some other Properties of Non-Metals ① They can be solid, liquids or gases at Room Temperature. ② They have low densities ie. they are light substances. ③ Non-Metals have many different colours: Sulphur - Yellow Phosphorus - White/Red Graphite - Black etc. Answers ① i) Mercury ii) Sodium iii) Copper and Aluminium iv) Lead ② Malleable means which is able to be hammered or pressed into shape without breaking or cracking. . A metal that can be drawn out into a thin wire is called ductile.

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Reaction of Metal with oxygen Metal + Oxygen → Metal Oxide * When metal are burnt in air, they react with oxygen in air to form metal oxide. -Metal oxides turn red litmus blue and they are basic in nature. $4\text{Na} + \text{O}_2(\text{g}) \rightarrow 2\text{Na}_2\text{O}(\text{s})$ metal Oxygen Sodium Oxide Aluminium burns in air, to form aluminium oxide. $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ Amphoteric Oxides Oxides that show basic as well as acidic behaviour are known as Amphoteric oxides. Aluminium Oxide is one of them. Zinc also forms amphoteric oxides. $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{HCl}_3 + 3\text{H}_2\text{O}$ Aluminium Chloride (salt) In this reaction, Al_2O_3 behaves as a basic oxide. $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_3 + \text{H}_2\text{O}$ Sodium Aluminate (salt) •Here Al_2O_3 , behaves as an acidic oxide.

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Reaction of metal with water: * When metal reacts with water (cold/hot) the product formed are metal hydroxide and hydrogen gas. Metal oxide + Water → Metal hydroxide •When a metal reacts with steam, then products are metal oxide and hydrogen gas. Metal + Steam → Metal oxide + Hydrogen
 $2K(s) + H_2O \rightarrow 2KOH + H_2 + \text{Heat}$ Water • The reaction is exothermic. •The reaction of Calcium with water is less violent. -The heat evolved is not sufficient for H gas to catch fire. $Ca + 2H_2O \rightarrow Ca(OH)_2(aq) + H_2(g)$ Calcium starts floating because bubbles of hydrogen gas stick to the surface of metals. • Metals like Al, Fe and Zn do not react with hot or cold water. • But they react with steam to form metal oxide and hydrogen [Also, lead, Gold, Silver → no reaction] ③Reaction of Metal with Acids : Metal + Dil. Acid → Metal + Hydrogen Salt * Magnesium reacts quite rapidly with dil. HCl acid forming magnesium chloride and hydrogen gas.

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Aqua Regia →It is freshly prepared mixture of 1 part of concentrated nitric acid and 3 parts of Concentrated hydrochloric acid. • Aqua regia can dissolve even gold and platinum metals but alone they can't do the same. (■■■■ ■■■■ ■■ ■■■■■■■■ ■■■■■■ ■■■!) ④Reaction of Metal with Salt solution : Salt Sol. of metal B + Metal A → Salt sol. of A + Metal B A more reactive metal displaces a less reactive metal from its salt solution. • Reaction of zinc with copper sulphate solution. $CuSO_4 + Zn \rightarrow ZnSO_4 + Cu$ Copper Sulphate Zinc Sulphate Reactivity Series • Arrangement of metals in a vertical column in order of decreasing reactivities is called REACTIVITY SERIES OF METALS. →Most reactive is placed at the top. →Least reactive is placed at the bottom.

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Sodium is a highly reactive element, that reacts with oxygen when comes in contact with air and burns. Therefore, it is kept immersed in kerosene for prevention. Iron with Steam: $3Fe_{(s)} + 4H_2O \rightarrow Fe_3O_4 + H_2$ a) Calcium with water: $Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + H_{2(g)}$ b) Potassium with water: $2K_{(s)} + 2H_2O_{(l)} \rightarrow 2KOH_{(aq)} + H_{2(g)} + \text{Heat}$ i) B-It gives displacement reaction with iron(II) sulphate. ii) Displacement reaction will take place, blue colour of copper(II) Sulphate solution will fade and red-brown deposit of copper will form on B. iii) B,A,C,D.

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Hydrogen gas is produced. → $FeSO_4 + H_2 \rightarrow Fe + H_2SO_4$ ⑤ When it is added in the solution, the colour of iron sulphate solution changes. - As zinc is more reactive than iron, so it displaces iron from its solution and a grey precipitate of iron and a colourless zinc sulphate is formed. $Zn + FeSO_4 \rightarrow ZnSO_4 + Fe$ How do metals and non-metals react? • When metals react with non-metals, they form ionic compounds and when non-metal reacts with a non-metal, they form covalent compounds. Causes of Chemical Bonding -An atom can achieve the inert gas electron arrangement in three ways: i) by losing one or more electron. ii) by gaining one or more electron. iii) by sharing one or more electron. IONS →An ion is an electrically charged atom. • It is formed by loss or gain of electrons by an atom. • So it contains an unequal number of electrons or protons. •+vely charged ion is known as cation. •-vely charged ion is known as anion. Ex: If Sodium and chlorine were to react $Na + Cl \rightarrow Na[Cl]$ Formation of Sodium Chloride

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→ Compounds having ionic bonds are known as Ionic bonds. •They are formed by transfer of electrons from one atom to another. •They are also known as Electrovalent Compounds. Properties of Tonic

Compounds ① Ionic compounds are usually crystalline solids because the oppositely charged ions attracts one another strongly and form a regular crystal. ② Ionic compounds have high melting and boiling point. ③ Ionic compounds are usually soluble in water but insoluble in organic solvent. ④ When we dissolve ionic solid in water or melt it, the crystal structure is broken down and ions become free to move and to conduct electricity.

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1 Mark Questions

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A7. It is due to the formation of basic copper carbonate $[\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2]$. (i). It does not react with water

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(d). (iii) and (iv) (d). Cu (c) Fe_3O_4

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3 Marks Questions decrease

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Calcium (20) – 2,8,8,2 Oxygen (8)-2,6 $\text{Ca} + \text{O} \rightarrow (\text{Ca}^{2+})[\text{:O}^{2-}]$

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green-coloured

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5 Marks Questions

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They conduct electricity in molten state as well as in aqueous solution. (i) Aluminium oxide is called an amphoteric oxide. (ii) An iron strip dipped in a blue copper sulphate solution turns the blue solution pale green. (iii) Hydrogen gas is not evolved when most metals react with nitric acid. (iv) Calcium does not occur in a free state in nature. (v) Sodium or potassium metals are kept immersed under kerosene. (i). Aluminium oxide is known as 'amphoteric oxide' due to its reaction with both acids and bases. (ii). The iron strip which dipped in the "blue copper sulphate solution" turned into the "blue solution pale green". This is due to the fact that there is "formation of iron sulphate" takes place inside the solution. (iii). When metals react with the nitric acid, it is oxidised by nitrate ion and not by hydrogen ion. (iv). Calcium doesn't occur in any 'free state' in nature, because it is "highly reactive in nature". (v). Sodium and potassium metals are reactive and act violently in presence of air. (a) (i) Write two properties of gold which make it the most suitable metal for ornaments.

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Silver and copper are the best conductors of heat. Gallium and cesium are the metals that melt when kept on palm.

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greenish black

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21. Assertion: Aluminium is more reactive than iron, its corrosion is less than that of iron Reason: Aluminium is covered with a strong protective layer of oxide which protects the metal from further corrosion. A. 21 (a) Both assertion (A) and reason (R) are correct and reason is the correct explanation of assertion.